

Kevin McLeod

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EDUCATION

Harvard University - Cambridge, MA

Expected May 2027

- A.B. in Computer Science and Physics (Double Major), GPA: 3.65/4.0
- **Relevant Coursework:** Distributed Computing, Systems Programming, Advanced Computer Networks, Data Structures and Algorithms, Machine Learning, Artificial Intelligence

SKILLS

Languages: C++, C, Python, Java, TypeScript, JavaScript, C#, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, FastAPI, Tailwind, Three.js / React Three Fiber, PyTorch, OpenCV, Unity

Infrastructure & Tools: PostgreSQL, Supabase, Celery, REST APIs, Git/GitHub, Linux, ROS

EXPERIENCE

Microsoft - Software Engineering Intern (C#, C++)

May – Aug 2026

- **Shipped a generative-AI image experience in Bing Images** end-to-end (Bing: **1B+ monthly users**), from design to review
- **Developed text-to-text quality metrics** for Bing image experiences and used them to improve multiple experiences by **7%+**
- Prototyped intent-driven, curated GenAI image experiences and a new routing design that serves stacked user needs

Stealth AI Startup - Co-founder & Engineering Lead (Next.js, React, FastAPI, Tailwind)

2025 – 2026

- **Architected the full stack from an empty repo to a working prototype:** Next.js/React frontend, FastAPI backend, AI models
- Ran discovery interviews and surveys with **100+ prospective users** and used the findings to narrow product scope

GP Enterprise Solutions - Full-Stack Software Engineering Intern (JavaScript, React, SQL)

Jun – Aug 2025

- Reduced database response times by **5%** by restructuring filter logic and eliminating redundant queries
- **Shipped bulk file export and dynamic form management features**, incorporating feedback from user interviews

Kempner Institute, Harvard - Research Assistant to Professor Yilun Du (PyTorch)

May – Aug 2025

- Estimated relative 3D object positions from single RGB frames using **Mask R-CNN** instance segmentation and **MiDaS** depth
- Built a PyTorch prototype composing per-object energy terms (ground contact, non-interpenetration, scene centering) into a **differentiable scene-level objective** minimized with Adam over 3D object positions

Harvard Datamatch - Web Developer (JavaScript, Next.js)

2024 – 2026

- Rebuilt the Datamatch frontend in Next.js on a 5-person team, serving **22,000+ students** during the matching window

PROJECTS

Collaborative Text Editor on Multi-Paxos - C++, JavaScript

[Source Code](#)

- Built a real-time collaborative editor in C++23 on **Multi-Paxos** with leader election, log trimming, and request batching
- Designed the editor state machine: splice-based edits, OT-style rebasing of stale edits, and dedup by client serial
- Reduced commit latency from **~370 ms to ~60 ms** under **1% packet loss** and **increased throughput by 43%**
- Added an HTTP/SSE gateway, binary protocol, and persistence with crash recovery; tested six failure scenarios

OS Kernel & Systems Programming - C, C++, x86-64

[Source Code](#)

- Implemented virtual memory, per-process page tables, and fork/exit/page-allocation syscalls in an x86-64 kernel
- Wrote a debugging malloc that catches double frees, invalid frees, out-of-bounds writes, and memory leaks
- Developed a Unix shell (pipes, redirection, conditionals, background jobs) and thread-safe I/O with byte-range locks

ClipV.I.S. - TypeScript, Three.js, Node.js, MediaPipe

[Source Code](#)

- Owned **real-time hand-gesture controls** (12 gestures: pinch-grab, two-hand scale, rotate, explode, snap views) from MediaPipe landmarks for a voice- and gesture-driven hologram, stabilized with hysteresis, debouncing, and EMA smoothing
- Integrated gesture recognition into a Three.js/React Three Fiber hologram; **placed 3rd in the Microsoft Intern Hackathon**

SLAM Agent for Maze Exploration - Python, NumPy

[Source Code](#)

- Implemented **Kalman-filter localization** and a log-odds occupancy grid from noisy simulated LIDAR raycasts
- Designed BFS frontier exploration with obstacle inflation and fallback replanning to map unknown mazes autonomously

Learned Structured Pruning During Fine-Tuning - Python, PyTorch

- Co-developed learned structured pruning for **BERT** on MMLU: trainable masks with sparsity and quantization losses
- Pruned **>90% of FFN activations** with a single-digit percentage-point accuracy drop vs. an unpruned baseline

Light in the Dark - C#, Unity

[Source Code](#)

- **Led a team of 10** building a 3D RPG, coordinating game design, audio, and UI work across disciplines
- Engineered animation state machines, modular combat, and Ink-based branching dialogue on an event-driven architecture

LEADERSHIP

Harvard Undergraduate Negotiation Club - Founding Board Member & Web Developer

Jan 2025 – Present

- Designed and launched the club's [website](#) with React and JavaScript